

ABHISHEK AGRAWAL

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Senior Software Engineer with 7+ years of experience in building scalable microservices and AI/ML solutions using **PyTorch**, **TensorFlow**, and **Big Data**. Skilled in developing **LLM-based applications** with **Langchain** and **RAG pipelines**. Proficient in **AWS**, **DevOps (Docker, Kubernetes, Terraform)**, and delivering real-world platforms

WORK EXPERIENCE

TECH INFOSKY PVT. LTD.

SR. Software Engineer

Nov 2024 – Present

- Designed smart AI tools that **understand and summarize large documents**, making it easier for teams to get insights without reading everything manually.
- Created systems where the AI can **look up answers from real documents or databases** before responding, using **RAG pipelines** for more reliable and factual outputs.
- Developed **microservices using FastAPI** in Python that handled different parts of big systems independently, so teams could work faster and apps could scale easily.
- Built **NLP models** to extract key info (like names, dates, or risks) from emails, reports, and messages to speed up data processing for business teams.
- Used **Elastic Cache** to temporarily store frequently-used data, helping apps respond faster and saving time during peak usage.
- Set up **Prometheus and Grafana** dashboards to monitor app performance, so the team could catch and fix issues before users noticed.
- Mentored junior developers, led code reviews, and made sure the team followed best practices throughout the **software development life cycle (SDLC)**.

ENTERPRISE MINDS (FLIPKART)

Software Engineer

Jul 2024 – Oct 2024

- Created **REST APIs** that let different apps and services securely talk to each other — like letting a dashboard fetch real-time updates or trigger actions with one click.
- Developed **modular backend services** with FastAPI, making the system easier to scale and maintain as the platform grew.
- Designed and optimized **MySQL databases** for fast and reliable storage of structured data, ensuring minimal latency during peak traffic.
- Automated server maintenance tasks using **Shell scripts on Unix systems**, saving hours of manual effort and reducing human errors.
- Managed cloud infrastructure on **AWS**, including **EC2 for compute**, **RDS for databases**, and **S3 for secure file storage**.
- Set up **auto-scaling and backup policies** to keep systems resilient, highly available, and cost-optimized.

CIPLA

Software Engineer

Aug 2022 – Jun 2024

- Created **REST APIs** that let different apps and services securely talk to each other — like letting a dashboard fetch real-time updates or trigger actions with one click.
- Built systems that **predict customer behavior or future trends**, helping businesses make smarter decisions (like when users might leave or what product they'll buy next).

- Used tools like **TensorFlow and PyTorch** to train models on large datasets, making predictions faster and more accurate with real-world impact.
- Built **automated alert systems** using **CloudWatch** and SNS to notify the team about unusual activity or system slowdowns.
- Designed dashboards using **AWS QuickSight** so business users could understand key metrics in a visual and interactive way without needing to query the database.
- Used **Ansible** to set up and configure multiple servers at once, saving hours of manual setup for each deployment.
- Worked in **Kanban and Scrum environments**, doing daily standups, planning sprints, and delivering working features in quick cycles.

NATCO PVT LTD

Associate Software Engineer

Sep 2017 – Jul 2022

- Developed **RESTful APIs using Python and FastAPI** to power real-time features in web and mobile apps, improving response times and scalability.
- Used **MySQL and PostgreSQL** to manage structured data, with proper indexing and query optimization for speed.
- Set up **role-based access controls (IAM)** so that only the right team members or services could access sensitive cloud resources.
- Deployed backend systems with **Docker and Jenkins**, automating the process from code to live production with minimal manual work.
- Collaborated in **Agile teams (Scrum/Kanban)**, participating in sprint planning, daily standups, and client demos to deliver features on time and with quality.

PROJECTS

AI-Powered Product Intelligence & Support System for E-commerce

Duration: Nov 2024 - Present

- Built an advanced **LLM-based product assistant and customer support solution** for a large e-commerce platform to enhance user experience, automate product discovery, and reduce support load by over 60%.
- Designed and deployed a **Langchain-powered conversational assistant** that helps users find the right products by understanding natural language queries like “Show me noise-cancelling headphones under ₹5000 with good reviews.”
- Integrated **RAG pipelines** to fetch real-time product data, specs, reviews, and FAQs from vectorized product databases and documents, improving accuracy and contextual relevance in responses.
- Used **OpenAI's GPT and Claude APIs** with customized prompts and dynamic chaining to handle follow-up questions, compare products, and explain features like a human expert.
- Developed a **multi-agent system** using **LangGraph** where different agents (Search Agent, Review Analyzer Agent, Comparison Agent) collaborate to fulfill complex customer queries seamlessly.
- Enabled **AI-powered customer support** for handling order tracking, return policies, and refund issues by integrating with backend systems and summarizing ticket logs using LLMs.
- Ensured smooth real-time performance using **FastAPI-based microservices**, deployed on **AWS (EC2, S3, RDS, DynamoDB)**, and monitored via **CloudWatch & Grafana dashboards**.
- Optimized prompt flows using Langchain callback handlers and tested AI response quality via **LangSmith**, reducing hallucinations by 35% and increasing customer satisfaction scores.

Technologies Used - Langchain, LangGraph, GPT-4, Claude, RAG, FastAPI, Python, AWS (S3, RDS, DynamoDB, EC2, CloudWatch), Vector DBs (FAISS, ChromaDB), Docker, CI/CD, Grafana, LangSmith.

Dynamic Pricing & Demand Forecasting System

Duration: Aug 2022 - Jun 2024

- Developed a real-time pricing intelligence system for a hotel chain to optimize room rates based on demand, competitor pricing, seasonality, and customer behavior.
- Built backend services using **Python (Django + Flask)** to manage rate adjustments, forecast data, and room availability APIs.
- Trained ML models with **PyTorch** for **demand forecasting and anomaly detection**, enabling dynamic price updates with up to **87% accuracy**.
- Implemented **predictive analytics** to forecast booking patterns and optimize promotions, increasing occupancy rate by **22%** during off-seasons.
- Stored historical and transactional data in **MongoDB and RDS**, processed large-scale booking and traffic logs using **Big Data pipelines**.
- Deployed microservices using **Docker and Kubernetes**, and automated infrastructure provisioning with **Terraform**.
- Integrated **AWS services (S3, RDS, SNS, SQS)** for data ingestion, storage, and event-based communication across internal systems.

Technologies Used - Python (Django, Flask), PyTorch, MongoDB, PostgreSQL (RDS), AWS (S3, SNS, SQS), Docker, Kubernetes, Terraform, Git, CloudWatch.

Guest Experience & Feedback Automation System

Duration: Sep 2017-Jul 2022

- Designed a **GenAI**-powered system to improve customer experience and automate feedback analysis across hotel and restaurant chains.
- Created **smart chat assistants using GenAI** to help guests with queries on amenities, bookings, check-ins, and recommendations.
- Used **NLP models in PyTorch** to analyze guest reviews, detect sentiment, and categorize feedback into actionable insights.
- Developed a **Flask-based backend** for processing real-time guest feedback, chatbot interactions, and escalation flows.
- Leveraged **anomaly detection algorithms** to identify service breakdowns or outlier behavior in staff/guest interactions.
- Processed unstructured feedback data with **MongoDB** and generated insights through **Big Data pipelines** integrated with **AWS S3 and DynamoDB**.
- Automated deployments and scaling using **Docker, Kubernetes, and Terraform** with continuous delivery pipelines.
- Enabled alerting and routing using **AWS SNS and SQS**, ensuring instant flagging of critical guest issues to the operations team.

Technologies Used - Python (Flask), Gen AI, PyTorch, NLP, MongoDB, DynamoDB, AWS (S3, SNS, SQS, IAM), Docker, Kubernetes, Terraform, Git, Grafana.

EDUCATION

ADDITIONAL

Skills: ~~LLM Models, NLP Models, RAG Pipelines, Langchain, Predictive Analysis, Prompt Engineering & Tuning, Python, FastAPI, RESTful APIs, Tensorflow, PyTorch, GenAI, SDLC, Microservices, MySQL, PostgreSQL, Linux OS, AWS, Quicksight, ElasticCache, SNS, SQS, DynamoDB, IAM, Cloudwatch, DevOps, Docker, Jenkins, Ansible, Git/Github, System design, Prometheus, Grafana, Kanban, SCRUM, Client Handling, Team Management.~~

Languages: English, Hind & Marathi.